Market Analysis

Indonesian Supermarket Business



Background & Objective

Background

The company needs a clear overview of its sales performance from 2014 to 2017 across time, regions, product categories, cities, and shipping methods.

Objective

To identify sales trends, regional and category contributions, the highest-performing city and month, and the most frequently used shipping method.

What was the trend in sales value from 2014 to 2017?

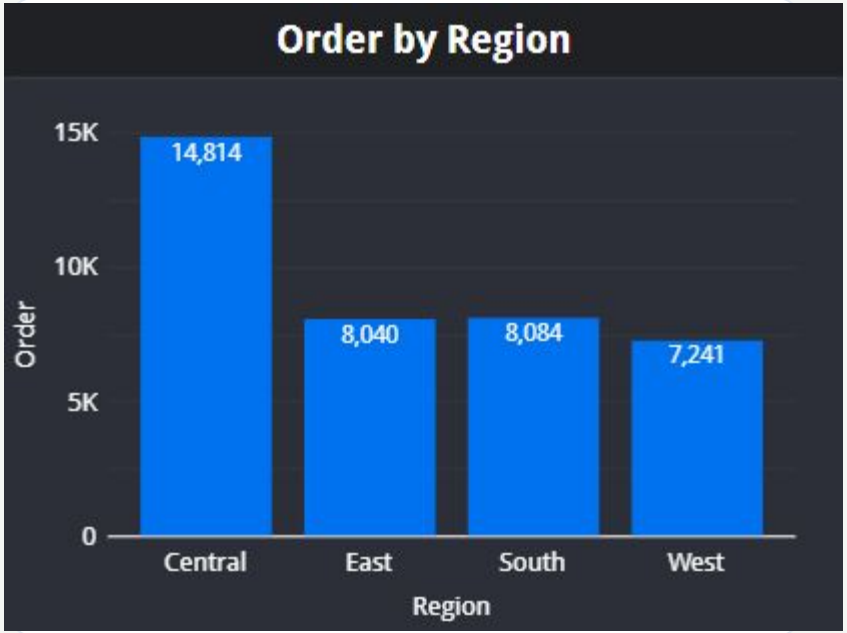


Consistent Growth Trend

+52% Overall Sales Increase

- **Initial Dip (2014-2015):** Sales started at 7.3B in 2014, experiencing a minor decline to 7.1B in 2015.
- **Strong Recovery (2016):** Sales rebounded sharply to 9.2B, showing a robust upward trajectory.
- **Peak Performance (2017):** Reached an all-time high of 11.1B, confirming strong market expansion.

What was the proportion of orders by region?



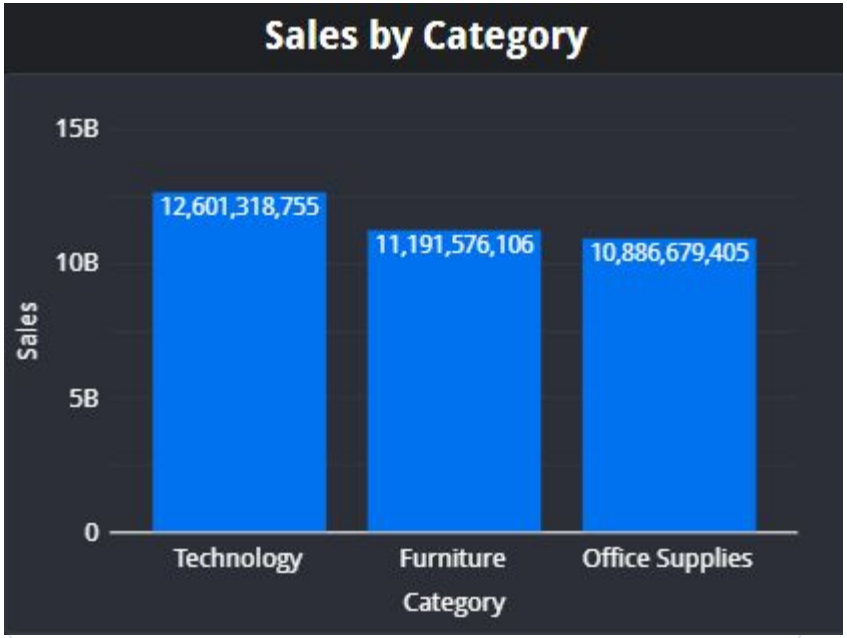
Regional Order Distribution

East & West

Drive Over 60% of Total Orders

- **East Region Leads:** Generates the highest proportion of supermarket orders.
- **West Region:** Follows closely as the second-largest contributor to sales volume.
- **Central & South:** Represent smaller, developing markets with high growth potential.

What was the sales proportion for each product category?



Product Category Sales

Technology Leads

With Over 12.6B in Total Sales

- **Technology:** Highest contributor with 12.6B, driven by high-value accessories.
- **Furniture:** Strong performance at 11.2B, showing consistent customer demand.
- **Office Supplies:** Close third at 10.9B, representing high-volume essential orders.

Which city generated the highest sales?



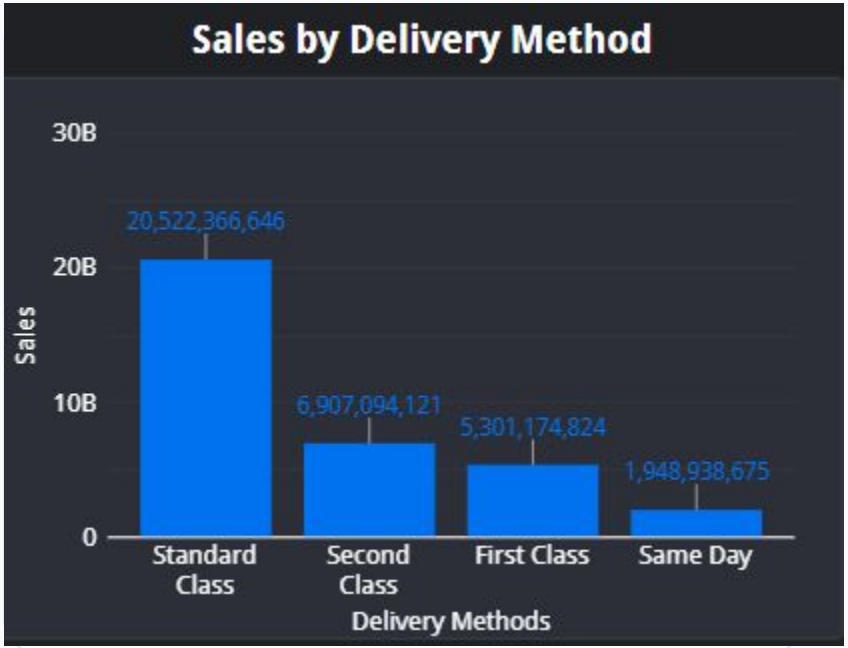
City Sales Performance

Balikpapan Leads

With Around Rp4.1 Billion in Total Sales

- **Highest Sales:** Balikpapan City generated the highest sales value.
- **Market Dominance:** Significantly outperformed other cities.

Which shipping method was most frequently used by buyers?



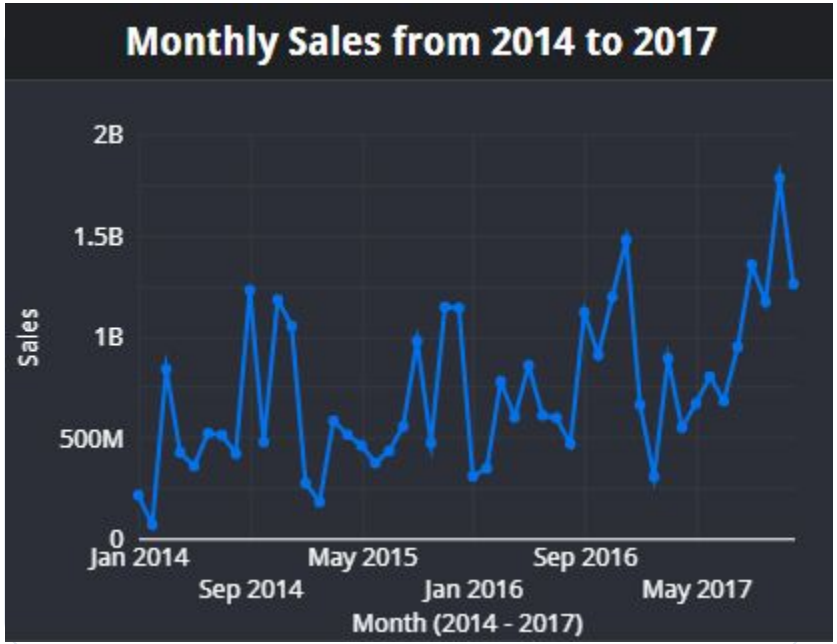
Shipping Method Preference

Standard Class

Dominates with Over 20.5B in Sales

- **Standard Class:** Highly preferred choice, representing the vast majority of orders.
- **Second & First Class:** Moderate usage, offering a reliable balance of cost and delivery speed.
- **Same Day:** Lowest frequency, typically selected only for urgent delivery requirements.

Which month saw the highest sales between 2014 and 2017?



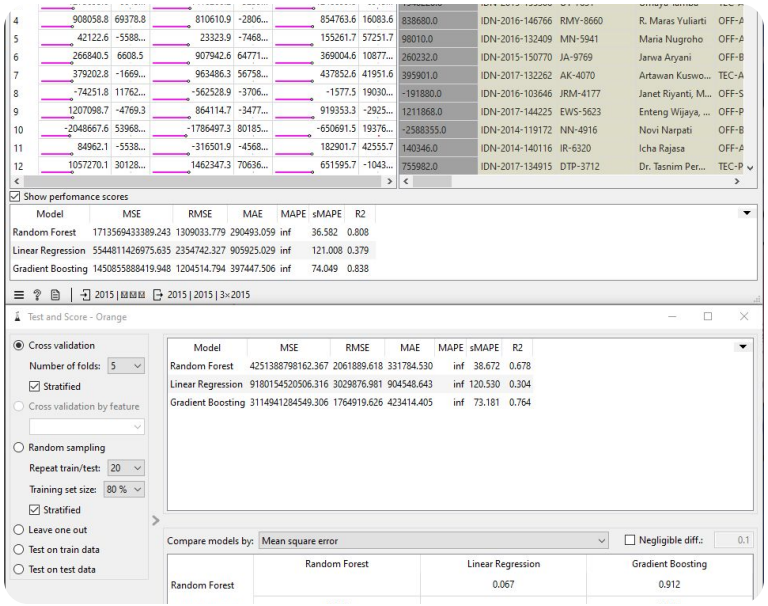
Sales Trend Analysis

November 2017

Peak Sales Reached Approximately IDR 1.8 Billion

- **Peak Sales:** Occurred in November 2017, reaching a value of approximately IDR 1.8 billion.
- **Growth Trend:** Shows seasonal fluctuations with a significant upward trend towards the end of 2017.

What is the estimated profit from the store's sales?



Model Performance Analysis

Gradient Boosting Wins

Selected as Primary Model with 0.838 R^2

- **Best Predictor:** Achieved lowest RMSE of **IDR 1.20 million** on holdout test data.
- **High Accuracy:** Explains **83.8%** of profit variation, beating Random Forest and Linear Regression.
- **Underprediction:** Estimated profit of **IDR 819.20M** vs. actual IDR 888.81M (7.83% error).

Conclusion & Recommendation

Key Conclusions

Tech & Balikpapan Drive Growth

- **Technology Dominance:** Technology leads category sales with 12.6B, followed closely by furniture.
- **Regional Stronghold:** Balikpapan emerges as the primary driver of regional sales performance.
- **Standard Shipping:** Standard Class remains the most preferred shipment choice for buyers.

Strategic Recommendations

Target Regional Tech Buyers

- **Optimize Product Mix:** Increase high-value accessory inventory in Tech to boost average order value.
- **Balikpapan Expansion:** Launch hyper-local marketing and build partnerships in high-performing regions.
- **Seasonal Campaigning:** Allocate promotional budgets heavily toward the peak November sales trend.